

FP8 training (V3): 8-bit for the big matmuls, higher precision where it matters

FP8 (8-bit): the heavy GEMMs

Fprop (forward matmul)

Dgrad (activation gradient)

Wgrad (weight gradient)

~2x compute throughput, less memory

Kept in BF16 / FP32: the sensitive parts

embedding & output head

MoE gating / router

normalization (RMSNorm)

attention

FP32 accumulation of FP8 GEMMs

Fine-grained scaling: 1x128 tiles (activations), 128x128 blocks (weights) -> loss error < 0.25% vs BF16